GymDash (Registry A): Phase 4.1 Master Spec

Architecture, Logic, and Operations Blueprint

Version 4.1.0 | Finalized

1. Core Infrastructure & Data Engine

GymDash Abandons the fractured microservice approach in favor of a single, highly optimized Supabase/Postgres schema. It is a live, event-driven engine where every physical action triggers a unified digital state. The goal is an airtight ecosystem where the front-desk tablet and the owner dashboard share one unbroken truth.

Append-Only Ledger

The financial core is immutable. Transactions are never edited or silently deleted. Voids, refunds, and SKU retirements are processed exclusively as reversing entries. This guarantees a flawless audit trail.

Real-Time Ambient State

The UI utilizes WebSocket subscriptions (not manual page refreshes). When a QR is scanned at the gate, the event ID instantly updates the tablet's Live Feed and bumps the "Inside Now" capacity counter in milliseconds.

Offline Resilience

If internet drops, the UI flashes "Offline." The scanner cryptographically verifies QRs against a locally cached active pass list. Cash sales queue locally with idempotency keys. Reconnection pushes logs to the server, routing discrepancies to the Manager.

Computational Caching

To support "All-Time" data views without database paralysis, the backend executes nightly cron jobs during off-peak hours. These aggregate lifetime gross sales, acquisitions, and cumulative churn into a cached dashboard view.

2. Identity, DTR, and Cash Accountability

To eliminate front-desk bottlenecks and enforce strict financial control, Human Resources (attendance) is structurally decoupled from the physical Cash Drawer.

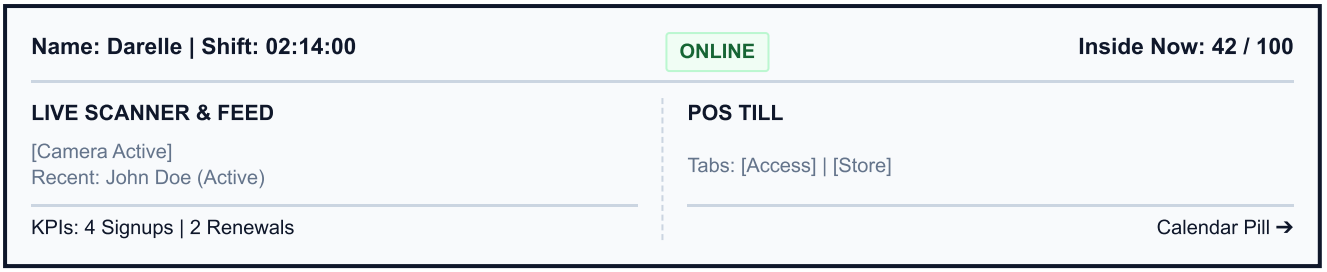
Mobile Selfie Clock-In (HR): Staff members do not clock in at the shared tablet. Upon arriving, they open their personal Sodia app (Staff Mode) and tap "Clock In." This captures a server-timestamped selfie, geofenced to the gym's physical coordinates. This proves on-site presence without stalling the lobby.

The Register Float Gate (Cash): Being "on the clock" does not grant register access. A staff member steps to the tablet, taps their name, and enters their PIN. The system strictly halts and demands the **Opening Cash Float**. Once physically counted and entered, all cash liability is bound to that staff member's shift ID until they hand over the drawer.

> **Mid-Shift Handover:** If Staff A takes a break, they tap "End Shift" (triggering a mid-shift blind count). Staff B then logs in with their PIN and assumes the drawer. The chain of financial custody never breaks, even though both remain clocked in via mobile.

3. The Daily Dash (Ambient UI Layout)

The Daily Dash is the operational floor for the staff, engineered for the "fast lobby" rule and aggressive cash accountability. It is the active resting state of the tablet once the shift float is declared.



4. Structural Layout Rules

Top Command Bar: Displays the logged-in staff name, shift duration, system connectivity, and the dynamic "Inside Now" estimated capacity. Staff uses this bar to initiate the Blind Count when ending a shift.

Left Panel (The Gate): Defaults to the active camera. It handles the Live Attendance Feed. Valid scans flash Green (Cleared) with the user's avatar. Denials flash Red. Below the feed are localized **Shift KPIs** (New Signups, Renewals, Expiring Today) to gamify the shift and prompt staff to pitch renewals.

Right Panel (The Till): The POS is strictly bifurcated. *Access* handles memberships and passes (subject to platform routing). *Store* handles physical goods (F&B/Merch) which route directly to the gym's cash or external settlements.

The Schedule Pill: A floating action button that opens a side-peek of the Daily Roster. Staff can instantly see who is on the clock via mobile, who is on the register, and which coaches currently hold active floor windows.

5. High-Speed User Flows (Front Desk)

Flow A: Live Scanner & Entry Overrides		
1. Scan	Member presents their 60-second dynamic QR code to the tablet. Backend checks validity, expiry, facility rules, and live capacity.	
2. Success	Screen flashes Green ("Cleared") with avatar. Event ID logged. "Inside Now" bumps +1.	
3. Denial	Screen flashes Red with specific reason code (e.g., "Pass Expired," "At Capacity").	
4. Override	Staff taps "Manager Override" to admit a denied scan. Staff CANNOT wave a user through silently.	
5. Approval	*Manager Present:* Enters PIN on tablet and selects reason code.	*Remote Approval:* Staff taps "Request." Push notification hits manager's phone for biometric unlock.

Flow B: POS Checkout & Unified Tendering	
1. Cart	Staff taps Access passes or Store items. Items queue into a single cart. Cart persists through errors.
2. Data Gate	Staff MUST tap "Who is buying?" Options: *App User, Walk-in, New Offline, Existing*. (Prevents anonymous data dumping).
3. Peak Rush	If line is long, a New Offline cash buyer is processed as a "Quick Cash / Walk-in." Receipt prints with a generated claim QR. User scans it later on their phone to complete profile via OTP.
4. Conflict	**Mixed Cart Tendering:** If cart contains an app-payable Access pass AND a gym-direct Store item, the "Pay via App" gateway dynamically disables. System forces a unified Cash or External Terminal payment for the entire cart to protect ledger routing.
5. Provision	System posts final state to the append-only ledger and issues entitlement.

Flow C: Corrections & Shift Close	
1. Voids	Staff selects flawed transaction in "Recent Sales" and taps Void/Refund. Screen locks demanding Manager PIN (local or remote). Reversing ledger row is posted.
2. End Shift	Staff taps "End Shift." The screen hides all mathematical totals.
3. Blind Count	Staff physically counts the drawer and types the total. System compares to expected (Float + Cash Sales - Refunds - Drops).
4. Review	If the count falls outside branch tolerance, shift is flagged. Staff must type an explanation, and the shift enters the Manager's review queue.

6. The Data Values & Entries Tool (SKU Engine)

This module governs the creation and pricing of all facility offerings. It is an infinite canvas—owners can indefinitely add, rename, and adjust prices for any product. However, to protect the database and the master ledger, every custom item is bound by strict, hard-coded categorical logic.

When an owner taps **"+ New Product,"** they navigate a rigid configuration flow to ensure the item maps perfectly to the front-desk POS and the financial reports.

Step	Logic & Action
1. Master Class *(Hard Dropdown)*	Owner MUST select the system type: • **Access:** Grants entry via QR. Subject to platform ledger rules. • **Store:** Physical goods. Bypasses digital entry; routes to gym cash.
2. Sub-Category	*If Access:* Daily Pass, Monthly Recurring, Fixed Term, Guest Pass. *If Store:* F&B, Apparel, Rentals (Lockers).
3. Definition	Custom Name (e.g., "Student Summer Promo"), Public Description, Base Price (Peso), Tax Flags, and Stock Count (Store only).
4. Publishing	Save as Draft, Schedule for later, or Publish immediately. Access SKUs instantly push to the public Sodia app facility profile.

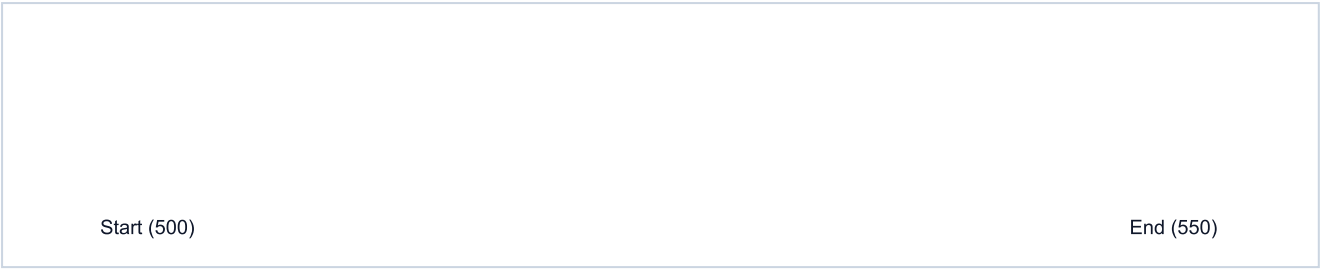
Immutability & Versioning Guardrails
Because the database relies on an append-only ledger, the tool enforces strict data integrity:
• **Versioning (Editing Values):** If an owner edits a "Basic Pass" from P1,000 to P1,200, the system does not overwrite the record. It generates "Basic Pass V2." New buyers pay V2; the system preserves V1 so existing members keep their locked-in contract terms.
• **Archiving (No Hard Deletes):** Owners cannot delete a product that has past transactions. They must "Archive" it, removing it from the POS but preserving all historical ledger math.

7. Manager/Owner Command Center

The dashboard abandons static spreadsheets in favor of active business intelligence. The absolute design rule is: **a metric with no attached action does not go on the home screen.** A master dropdown (Daily, Weekly, Monthly, Quarterly, Annual, All-Time) dynamically recalculates the entire interface.

Growth: The Member Lifecycle Equation

The Growth tab visualizes the facility's exact trajectory via a Net Equation. Tapping the ending total opens the SKU Mix report to show tier distribution.



Home ("Today"): The Triage Engine

The Home tab eliminates data-hunting by ranking operational fires. The "Needs Attention" queue flags blind-count mismatches, pending remote overrides, and capacity warnings. Each item carries direct "Approve," "Recount," or "Escalate" buttons. The top strip monitors live revenue and peak capacity.

Members: Predictive Attrition CRM

Because frequency drops months before cancellation, the system actively intercepts churn. Members expiring within 7 days, or absent for 14 days, are grouped into the "At-Risk" funnel. Managers sort this queue by Visit Frequency. A single tap on "Send Offer" pushes a discounted renewal SKU directly to the selected members' Sodia inbox.

Money: Ledger Control & Yield Management

This tab secures the cash and maximizes floor utilization. It features the master ledger, platform fees, and net payouts. Two critical algorithms operate here:

- **Gate-versus-Till Gap:** The system automatically calculates unrecorded cash by subtracting logged anonymous sales from physical anonymous admits, instantly flagging staff skimming.
- **Hourly Heatmap:** Tracks the "inside-now" headcount by hour. When dead-hour thresholds are met, the system prompts the owner to use the SKU engine to deploy Off-Peak passes.

Team: Roster & Coach Affiliation

A unified toggle switches between the Live Roster (showing mobile selfie status and active register assignments) and the macro Calendar grid for scaling shifts. Managers invite independent coaches via a Sodia handshake; the coach sets their floor windows, and the gym tracks their client volume. The gym retains absolute rights to pause or revoke a coach's access instantly.

Settings: The Data Import Wizard (Transition Tool)

To onboard gyms migrating from paper or spreadsheets, GymDash features a non-disruptive CSV import tool. Owners upload their legacy lists (Name, Phone, Pass Expiry). The backend scripts these as active `offline_members`. As these legacy users physically arrive at the gym, the front-desk verifies their phone number and issues the digital claim code to link their legacy history to the Sodia ecosystem.

Macro Valuation Metrics (All-Time View)	
GetPhysical Value Card	Isolates cumulative app-sourced acquisitions and revenue to definitively prove the platform's lifetime ROI to the gym owner.
Business Valuation	Generates cumulative lifetime Gross Sales and True LTV (Lifetime Value) required for banking, expansion, or sale of the business.